

PROJECT DEVELOPMENT REPORT

Vibe Coding & AWS Cloud Deployment | IBM & Bharat Cares

Live Application URL: <https://15.135.202.64.sslip.io>

1. Application Overview & Tech Stack

StudyMate.AI is a full-stack, AI-powered study platform built using modern Vibe Coding techniques and deployed on Amazon Web Services (AWS).

Tech Stack Breakdown:

- Frontend: React 19, Vite, Tailwind CSS v4, Motion animations, Lucide React icons.
- Backend: Node.js 20, Express.js framework, Esbuild bundler, Mammoth document parser.
- AI Model: Google Gemini 3.6 Flash (@google/genai SDK).
- Cloud Infrastructure: AWS EC2 (t2.micro / Ubuntu 24.04 LTS), Nginx Reverse Proxy, PM2 Process Manager, Let's Encrypt Certbot SSL.

2. Prompting Strategy & Frameworks Used

Persona & System Prompting

The AI is assigned a pedagogical persona ('Expert Academic Tutor') instructed to maintain clarity, encouraging tone, and structured formatting.

Structured Output Schema (JSON)

For Quiz Generation, prompts enforce a rigid JSON schema enforcing question text, options array, correct option index, and detailed explanation to ensure flawless UI parsing.

Sample System Prompt Used:

"You are StudyMate AI, an expert exam tutor. Given the study notes below, generate a 5-question multiple choice quiz. Output ONLY raw JSON matching this schema: [{ id, question, options: [4 strings], correctIndex: 0-3, explanation }]."

3. Phase-by-Phase Development Summary

Phase 1: Conceptualization & UI Design

Designed a frosted-glass UI system with dark/light themes, responsive navigation, and tabbed workflow (Notes, Tutor, Quiz).

Phase 2: Backend & LLM Integration

Developed Express API endpoints (/api/simplify, /api/tutor, /api/quiz) ensuring Gemini API keys remain strictly secure on the server side.

Phase 3: Codebase Hardening & Git Workflow

Configured strict .gitignore rules to prevent credential leaks, pushing clean code to GitHub.

Phase 4: AWS Cloud Infrastructure Setup

Provisioned an AWS EC2 t2.micro instance, configured Security Groups (SSH, HTTP, HTTPS), installed Node.js 20, PM2, and Nginx.

Phase 5: SSL Security & Responsive Polish

Configured Nginx reverse proxy with Let's Encrypt Certbot on 15.135.202.64.sslip.io and optimized mobile viewport layout.

4. Application Architecture

Client Browser % % (HTTPS/Port 443)% % %° Nginx Reverse Proxy % % (Port 80 to 3000)% % %° PM2 Node.js Express SDK)% % %° Google Gemini 3.6 API

5. Challenges Encountered & Resolutions

Challenge 1: GitHub Push Protection Block

GitHub blocked initial push due to API key detection in studyAI.env. Resolved by resetting git history, configuring .gitignore (*.env), and managing secrets via server environment variables.

Challenge 2: Vite Build Entry Path Resolution

Vite build failed to locate index.html on server root. Resolved by running 'npx vite build frontend --outDir ../dist'.

Challenge 3: PM2 Runtime & Dependency Resolution

Server threw 'Cannot find module mammoth' on startup. Resolved by installing mammoth package globally/locally and starting PM2 with NODE_ENV=production.

Challenge 4: HTTPS SSL Setup on AWS IP

IP addresses cannot obtain standard SSL certs. Resolved by using 15.135.202.64.sslip.io with Let's Encrypt Certbot.

6. Key Learnings & Reflection

This project demonstrated the power of Vibe Coding—using AI assistance for accelerated full-stack development, server management, and cloud deployment. The project achieved a 100% functional, secure, and production-ready application on AWS Free Tier.